lecture-07

Vector space, $(V, \oplus, \odot, \mathbb{F})$

Set

Addition

scalar

Linear Combination

Suppose, $S = \{\underbrace{v_1, v_2, v_3, \ldots, v_n}\}$ elements of this set belongs to V vector space.

this is merely a set

Let's pick, $\underbrace{v_{fix}}_{\text{vector}} \in V$.

Motivation

$S = \left\{ \underset{v_1}{\begin{pmatrix} 1 \\ 0 \end{pmatrix}}, \underset{v_2}{\begin{pmatrix} 0 \\ 1 \end{pmatrix}} \right\}$ $\quad V = \mathbb{R}^2$

Any random, $\begin{pmatrix} 5 \\ 6 \end{pmatrix} \in V$

$$5\begin{pmatrix} 1 \\ 0 \end{pmatrix} + 6\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 5 \\ 6 \end{pmatrix}$$

$S = \left\{ \underbrace{\begin{pmatrix} 1 \\ 1 \end{pmatrix}}_{v_1}, \underbrace{\begin{pmatrix} 0 \\ 1 \end{pmatrix}}_{v_2} \right\}$

Unknown

$$\underbrace{\alpha v_1 + \beta v_2 = v_{fix}}_{\text{linear combination}}$$

$$\alpha \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \beta \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 5 \\ 6 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} \alpha \\ \alpha \end{pmatrix} + \begin{pmatrix} 0 \\ \beta \end{pmatrix} = \begin{pmatrix} 5 \\ 6 \end{pmatrix}$$

$$\left.\begin{aligned} \alpha + 0 &= 5 \\ \alpha + \beta &= 6 \end{aligned}\right\}$$ By inspection $\alpha = 5$, $\beta = 1$

$$\left(\begin{array}{cc|c} 1 & 0 & 5 \\ 1 & 1 & 6 \end{array}\right) \sim \quad - - - \quad \alpha = 5, \beta = 1.$$

<u>Example</u>:

$$\underbrace{M_{2\times 2}}(\mathbb{R}) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \middle| \underbrace{a, b, c, d \in \mathbb{R}} \right\}$$

position

$$\begin{pmatrix} \boxed{A_{11}} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}$$

$$k \cdot \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} ka & kb \\ kc & kd \end{pmatrix}$$

$$\underbrace{e_{11} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}}, \quad \underbrace{e_{12} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}}, \quad \underbrace{e_{21} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}}, \quad \underbrace{e_{22} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}}$$

$$V_{fix} = \begin{pmatrix} 5 & 1 \\ -2 & 3 \end{pmatrix}$$

$$5e_{11} + 1e_{12} + (-2)e_{21} + 3e_{22} = V_{fix}$$

$$e = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix} \quad f = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} \quad g = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

$$\alpha\begin{pmatrix}1&0\\1&0\end{pmatrix}+\beta\begin{pmatrix}1&-1\\0&1\end{pmatrix}+\gamma\begin{pmatrix}0&1\\-1&0\end{pmatrix}=\begin{pmatrix}5&1\\-2&3\end{pmatrix}$$

$$\Rightarrow\begin{pmatrix}\alpha+\beta+0\cdot\gamma & 0\alpha-\beta+\gamma\\ \alpha+0\cdot\beta-1\gamma & 0\alpha+\beta+0\cdot\gamma\end{pmatrix}=\begin{pmatrix}5&1\\-2&3\end{pmatrix}$$

$$\left.\begin{aligned}\alpha\quad\alpha+\beta&=5\\ -\beta+\gamma&=1\\ \alpha-\gamma&=-2\\ \beta&=3\end{aligned}\right\}$$

2 choice

$$\left(\begin{array}{ccc|c}1&1&0&5\\0&-1&1&1\\1&0&-1&-2\\0&1&0&3\end{array}\right)$$

$$\sim\left(\begin{array}{ccc|c}1&1&0&5\\0&-1&1&1\\0&-1&-1&-7\\0&1&0&3\end{array}\right)\sim\left(\begin{array}{ccc|c}1&1&0&5\\0&1&-1&-1\\0&-1&-1&-7\\0&1&0&3\end{array}\right)\quad R_2'=-R_2$$

$$\sim\left(\begin{array}{ccc|c}1&1&0&5\\0&1&-1&-1\\0&0&-2&-8\\0&0&1&4\end{array}\right)\sim\left(\begin{array}{ccc|c}1&1&0&5\\0&1&-1&-1\\0&0&1&4\\0&0&1&4\end{array}\right)\quad R_3'=\frac{R_3}{-2}$$

You have two choice which gives $\gamma=4$

$$\begin{array}{c}\begin{array}{ccc}\alpha&\beta&\gamma\end{array}\\ \sim\left(\begin{array}{ccc|c}1&1&0&5\\0&1&-1&-1\\0&0&1&4\\0&0&0&0\end{array}\right)\end{array}$$

$\gamma=4$

$\beta-\gamma=-1\Rightarrow\beta=-1+4=3$

$\alpha+\beta=5\Rightarrow\alpha=5-\beta=5-3=2$

Sidenote:

$$\begin{array}{c}\begin{array}{ccc}\alpha&\beta&\gamma\end{array}\\ \left(\begin{array}{ccc|c}1&1&0&5\\0&1&-1&-1\\0&0&0&0\end{array}\right)\end{array}$$

free = # variables − # pivots
= 3 − 2 = 1

or

$$\left(\begin{array}{ccc|c}1&1&0&5\\0&1&-1&-1\\0&0&0&1\end{array}\right)$$

No Solution

Example

$$v_1 = (1,-2,1), \quad v_2 = (2,1,-1), \quad v_3 = (7,-4,1)$$

$$v = (6,-7,3)$$

Mandatory

$$\boxed{\alpha v_1 + \beta v_2 + \gamma v_3 = v}$$

$$(\alpha,-2\alpha,\alpha) + (2\beta,\beta,-\beta) + (7\gamma,-4\gamma,\gamma) = (6,-7,3)$$

$$(\alpha+2\beta+7\gamma,\ -2\alpha+\beta-4\gamma,\ \alpha-\beta+\gamma) = (6,-7,3)$$

$$\left.\begin{aligned} \alpha+2\beta+7\gamma &= 6 \\ -2\alpha+\beta-4\gamma &= -7 \\ \alpha-\beta+\gamma &= 3 \end{aligned}\right\} \qquad \left(\begin{array}{ccc|c} 1 & 2 & 7 & 6 \\ -2 & 1 & -4 & -7 \\ 1 & -1 & 1 & 3 \end{array}\right)$$

Try yourself

$$\alpha = 4-3t$$

$$\beta = 1-2t$$

$$\underline{\gamma = t}$$

$$\left(\begin{array}{ccc|c} & & & \\ & & & \\ 0 & 0 & 0 & 0 \end{array}\right)$$

$t \in \mathbb{R}$

$$\underbrace{(4-3t)v_1 + (1-2t)v_2 + t\,v_3}_{} = (6,-7,3)$$

Unique/Many : Linear combination exists

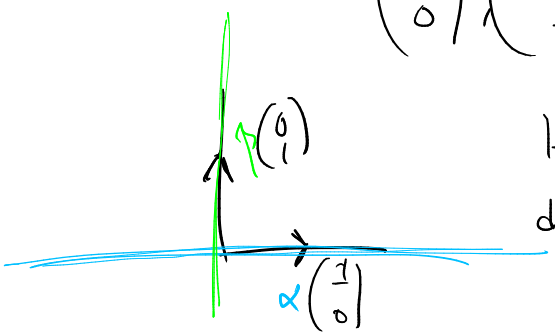
No Solution : No such linear combination exist

Linear Independence

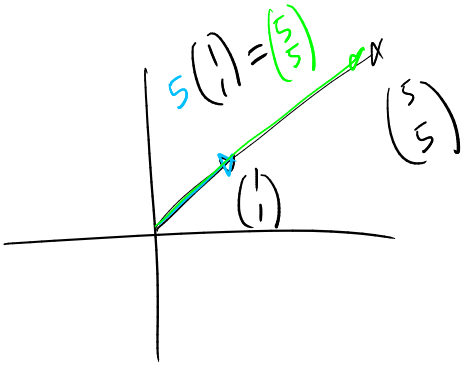
$$\underbrace{S}_{\text{set}} = \{\underbrace{v_1, v_2, \ldots v_n}\}, \quad \text{elements belong to } V \text{ (vector space)}$$

Motivation :

$$\begin{pmatrix}1\\0\end{pmatrix}, \begin{pmatrix}0\\1\end{pmatrix}$$



Here they represent unique direction. (Independent)



Dependent

$$\boxed{\begin{array}{l} \mathbb{R}^3 : \vec{0} = (0,0,0) \\ M_{2\times2} : \vec{0} = \begin{pmatrix}0&0\\0&0\end{pmatrix} \end{array}}$$

Mathematical Motivation: v_1, v_2, v_3

$$\alpha v_1 + \beta v_2 + \gamma v_3 = \vec{0}$$

Assumption, α, β, γ are no zero,

$$\alpha v_1 = -\beta v_2 - \gamma v_3$$

$\alpha = 0$,

$$\boxed{v_1 = \frac{1}{\alpha}(-\beta v_2 - \gamma v_3)}$$

$\beta \neq 0,$ $\boxed{v_2 = \frac{1}{\beta}\left(-\alpha v_1 - \gamma v_3\right)}$

$\gamma \neq 0,$ $\boxed{v_3 = \frac{1}{\gamma}\left(-\alpha v_1 - \beta v_2\right)}$

Formula:

$$\alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3 + \cdots \alpha_n v_n = \vec{0} \text{ (zero vector)}$$

IF $\boxed{\alpha_1 = \alpha_2 = \cdots = \alpha_n = 0}$

then $v_1, \ldots, v_n$ are Independent

Example:

$$v_1 = (2,1,2), \quad v_2 = (0,1,-1), \quad v_3 = (4,3,4)$$

$v_1, v_2, v_3 \in \mathbb{R}^3$

$$\alpha v_1 + \beta v_2 + \gamma v_3 = \vec{0}$$

$$\Rightarrow (2\alpha, \alpha, 2\alpha) + (0, \beta, -\beta) + (4\gamma, 3\gamma, 4\gamma) = (0,0,0)$$

$$\left.\begin{aligned} 2\alpha + 4\gamma &= 0 \\ \alpha + \beta + 3\gamma &= 0 \\ 2\alpha - \beta + 4\gamma &= 0 \end{aligned}\right\}$$

$R_1 = \frac{R_1}{2}$

$$\left(\begin{array}{ccc|c} 2 & 0 & 4 & 0 \\ 1 & 1 & 3 & 0 \\ 2 & -1 & 4 & 0 \end{array}\right) \sim \left(\begin{array}{ccc|c} 1 & 0 & 2 & 0 \\ 1 & 1 & 3 & 0 \\ 2 & -1 & 4 & 0 \end{array}\right) \sim \left(\begin{array}{ccc|c} 1 & 0 & 2 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & -1 & 0 & 0 \end{array}\right)$$

$$\sim \left(\begin{array}{ccc|c} 1 & 0 & 2 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right)$$

$\gamma = 0, \quad \beta + \gamma = 0 \Rightarrow \beta = 0,$

$\alpha + 2\gamma = 0 \Rightarrow \alpha = 0$

Unique: $(\alpha=0, \beta=0 \ldots)$ Independent

Many/No Solution: Dependent

Example:

$$v_1 = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \quad v_2 = \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix} \quad v_3 = \begin{pmatrix} 7 \\ 8 \\ 9 \end{pmatrix}$$

$$v_2 = \overset{\alpha}{\frac{1}{2}} v_1 + \overset{\beta}{\frac{1}{2}} v_3$$

(a)

$$\alpha v_1 + \beta v_2 + \gamma v_3 = \vec{0}$$

$$R_2' = \frac{R_2}{-3} \qquad R_3' = \frac{R_3}{-6}$$

$$\left(\begin{array}{ccc|c} 1 & 4 & 7 & 0 \\ 2 & 5 & 8 & 0 \\ 3 & 6 & 9 & 0 \end{array}\right) \sim \left(\begin{array}{ccc|c} 1 & 4 & 7 & 0 \\ 0 & -3 & -6 & 0 \\ 0 & -6 & -12 & 0 \end{array}\right) \sim \left(\begin{array}{ccc|c} 1 & 4 & 7 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 1 & 2 & 0 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & 4 & 7 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

Because our system has many solution, it's not independent.

(N.B.: No need to find out parametric solution)

(b)

$$\boxed{\begin{array}{l} \alpha v_1 + \beta v_2 = v_3 \\ \text{(OR)} \quad \alpha v_2 + \beta v_3 = v_1 \\ \text{(OR)} \quad \alpha v_1 + \beta v_3 = v_2 \end{array}}$$

linear combination

third.

Example

$$v_1 = \begin{pmatrix} 6 \\ 2 \\ -1 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 6 \\ 2 \\ -1 \end{pmatrix}$$

$$\alpha v_1 + \beta v_2 = \vec{0}$$

$$\left(\begin{array}{ccc|c} 0 & 0 & 0 & 0 \end{array}\right)$$